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Technical Report

Design space definition for a stirred single-use bioreactor family from 50 to 2000 L scale

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Single-use bioreactors continue to gain large interest in the biopharmaceutical industry. They are frequently used for mammalian cell cultivations, e.g. production of monoclonal antibodies and vaccines. This is motivated by several advantages of these bioreactors such as reduced risk of cross contaminations or short lead times. Single-use bioreactors differ in terms of shape, agitation principle, and gassing strategy. Hence, a direct process transfer or scale-up can be a challenge. Conventional stirred tank bioreactor designs are therefore still considered as the gold standard due to their well-defined and characterized properties. Based on this knowledge, a stirred single-use bioreactor family from 50 to 2000 L scale was developed with geometrical ratios similar to conventional reusable systems. To follow a quality by design approach, the single-use bioreactor family evaluated here was characterized by using process engineering methods such as the power input per volume, the mixing time, and the $k_L a$ value. The process engineering characterization demonstrates that these systems are suitable for cultivations of mammalian cells, even for high cell density and high titer applications. Based on the data, a scale-up or process transfer is possible between this bioreactor family as well as reusable systems. Therefore, this bioreactor family represents a major progress for the single-use technology.

Keywords: Design space / Process engineering characterization / Scale-up / Quality by design / Stirred single-use bioreactor

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1 Introduction

For biopharmaceutical processes, the used bioreactor as well as its design is of major importance. The bioreactor is a key element for process optimization [1] as operations such as homogenization, dispersion, suspension of solid particles, and heat transfer [2] are performed. Due to the well-defined characteristics of stirred stainless steel and glass bioreactors, these cultivation platforms are the gold standard since many decades for biopharmaceutical requirements.

During the last years, single-use bioreactors have gained more and more acceptance in the field of biotechnology. This is motivated by several advantages these bioreactors offer, such as reduced operational and investment costs. Furthermore, the time to market is shortened by lower lead times and reduced qualification efforts [3, 4]. As a consequence, single-use technology

is an answer to the increasing cost pressure in the industry for monoclonal antibody and vaccine production [5].

Many processes are carried out in stirred bioreactors where mixing and power input are achieved by – usually multiple – impellers. These systems are in general actively aerated. The design of most single-use bioreactors differs from conventional stainless steel or glass fermenters [6, 7] making a scale-up and process transfer to other systems challenging. To circumvent these difficulties, Sartorius Stedim Biotech GmbH developed the bioreactor family BIOSTAT® STR, which has the same well-known design criteria of reusable bioreactors [8].

The bioreactor needs to fulfill the specific process requirements of the respective cultivation. As a consequence, a detailed knowledge of the used cultivation system is necessary for a scale-up or process transfers [9], which can be obtained by process engineering characterization.

1.1 Process engineering characterization

Mammalian cells are sensitive against shear forces because they do not have a cell wall. The shear forces are proportional to the power input per volume (P/V_L) [7], a generally used scale-up criterion. It describes the transferred energy of the agitation

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Abbreviations: CHO, Chinese hamster ovary; PBS, phosphate buffered saline; QbD, quality by design; RO, reverse osmosis; STR, stirred tank reactor

device into the medium and correlates with the ability to suspend solids, disperse gas bubbles, and mixing behavior of a bioreactor [2]. For cell culture processes P/V_L between 10 and 250 W m⁻³ have been reported from lab to production scale [7]. For the description of the power input, the Newton number (Ne) is essential. With this value, conclusions about the efficiency of an impeller configuration can be made [2]. By knowing Ne of a bioreactor, P/V_L can be calculated.

One purpose of an agitation device is the homogenization, necessary to compensate concentration and temperature gradients. These gradients can have disadvantage effects on the cell growth and protein expression [10]. For its quantification, the mixing time can be used. They depend on the reactor geometry, impeller design as well as P/V_L [11]. The mixing time defines the time needed to completely homogenize a solution after the occurrence of a concentration difference. Mixing times below 60 s are described for the cultivation of mammalian cells to avoid disadvantage effects on the cell growth [10]. Different methods are established for the characterization of the mixing times [7]. In this report, the decolorization method was preferred, which is based on the color change of a dye.

Oxygen is an essential nutrient for aerobic processes and is consumed by the cells for growth, maintenance, and metabolic production [12]. The oxygen demand is described by the oxygen uptake rate (OUR [mmol/(L·h)], see Eq. (1)). The OUR correlates with the cell density (X [cells L⁻¹]), specific growth rate (μ [h⁻¹]), and the oxygen yield coefficient (Y_{X/O_2} [cells/mmole_{O₂}]), which depends on the used organism.

$$OUR = \frac{\mu}{Y_{X/O_2}} \cdot X \quad (1)$$

To ensure a sufficient supply of oxygen a permanent transfer to the medium by e.g. sparger has to be facilitated. The transfer of oxygen from gas bubbles into the medium is defined by the oxygen transfer rate (OTR [mmol L⁻¹ h⁻¹]) (see Eq. (2)). It depends on the difference of the oxygen saturation concentration (C^* [mmol L⁻¹]) and the dissolved oxygen concentration (C [mmol L⁻¹]) defining the driving force [12]. Furthermore, it is affected by the volumetric mass transfer coefficient ($k_L a$ [h⁻¹]) being the reciprocal time needed for the oxygen transfer from the gas bubble to the liquid phase. This value is influenced by the bioreactor and impeller design as well as P/V_L and the gassing strategy [12]. For most mammalian cell cultivations, a $k_L a$ of ~ 10 h⁻¹ is sufficient [6].

$$OTR = k_L a \cdot (C^* - C) \quad (2)$$

1.2 Scale-up

After a cultivation process is established on laboratory scale, the aim is to transfer the process into larger bioreactors (scale-up). Deviations can occur during a process scale-up because mass transfer phenomena are highly dependent on the scale and the presence of spatial gradients [10].

A basic prerequisite for a successful scale-up is the geometrical similarity (e.g. height to diameter ratio, impeller to vessel diameter ratio) [13]. Furthermore, the material properties of the medium such as the pH, temperature, and composition have to remain the same. It is impossible to keep all process engineering

parameters constant during a scale-up. Hence, a relevant parameter has to be defined, which remains constant. A commonly used scale-up criterion is to maintain P/V_L [9] or for sheer sensitive cells, the tip speed [14].

Based on a detailed knowledge from reusable bioreactors, important design criteria can be given for the cultivation of mammalian cells. An aspect often considered is the height to vessel diameter ratio (H:D), which varies between 1:1 and 3:1 for stirred tank reactors [13]. The ratio has an influence on the gas transfer because it correlates with residence time of gas bubbles for bioreactors with direct sparging [15]. For the cultivation of mammalian cells, a ratio of 2:1 is recommended [16].

A further parameter is the impeller to vessel diameter ratio having an effect on the mixing efficiency and the generated shear rates. For cell culture applications, the ratio should be between 0.33 and 0.5 [17]. For Newtonian culture broths like mammalian cell suspensions often segment impellers are used to achieve mixing generating an axial flow pattern. Segment impellers transform the transferred energy efficient into hydrodynamic power and generate large circulation loops. This correlates with a suitable mixing behavior, and therefore they are preferred for homogenization. Furthermore the transferred dissipative energy is low, and consequently they are often used for sheer sensitive cells. However, in the past decade, optimization of different cell strains resulted in less sheer sensitive cells. This also allows the usage of a disk impeller on the bottom and a segment impeller on the top of the stirrer shaft. In this way, the efficient mixing performance of the segment impeller gets combined with the enhanced bubble distribution of the disk impeller, making this configuration interesting for high density applications [18, 19].

It is common to install multiple impellers in bioreactors with a H:D ratio above 1:1.4 to ensure an efficient mixing through the entire cultivation chamber. A distance between the impellers of 1.2–1.5 times the impeller diameter (d_2) should be selected because the impellers act independent of each other.

Besides the mixing efficiency, the gas transfer is crucial for the design of a bioreactor. For the aeration, generally ring spargers with drilled holes are in general installed below the impellers [20]. The gas transfer is influenced by the size and amount of generated gas bubbles. Smaller bubbles have a higher specific transfer area (a), which significantly improves the overall gas transfer [12]. On the other hand, a disadvantage is that cells can be damaged when bubbles burst. Bubble sizes are generally in the region of 4–6 mm. Larger bubbles have the advantage that CO₂-stripping is enhanced, and consequently an accumulation in the cultivation medium can be reduced.

2 Materials and methods

2.1 Design criteria for the CultuBag STR

To overcome the existing challenges during a process scale-up, a stirred single-use bioreactor family (BIOSTAT® STR, Sartorius Stedim Biotech GmbH, Germany) was developed based on the proven and trusted principles of stainless steel fermenters [21]. The main application of this bioreactor family is the cultivation of mammalian cells and is available from 50 to 1000 L scale, while a 2000 L scale is under development. These systems have a

Table 1. Summary of key geometrical dimensions^{a)} for the CultiBag STR family [8].

CultiBag STR	50	200	500	1000	2000
Total volume [L]	68	280	700	1300	2800
Max. working volume [L]	50	200	500	1000	2000
Min. working volume [L]	12.5	50	125	250	500
Bag diameter D [mm]	370	585	815	997	1295
Bag height H [mm]	666	1055	1467	1800	2330
Ratio H:D [–]	1.8	1.8	1.8	1.8	1.8
Liquid height h_1 [mm]	480	783	1005	1360	1670
Ratio h_1/D [–]	1.3	1.34	1.23	1.36	1.29
Impeller diameter d_2 [mm]	143	225	310	379	492
Ratio d_2/D [–]	0.38	0.38	0.38	0.38	0.38
Distance between impellers Δz [mm]	186	300	403	493	640
Size drilled holes ring sparger part [mm]	0.8	0.8	0.8	0.8	0.8
Number drilled holes ring sparger part [–]	5	25	100	100	200
Size drilled holes micro sparger part [mm]	0.15	0.15	0.15	0.15	0.15
Number drilled holes micro sparger part [–]	25	100	500	500	1000

^{a)} Key geometrical dimensions are represented in Fig. 1.

single-use cultivation chamber (bag) made of synthetic materials. The bags have a cylindrical shape with a H:D ratio of 2:1, an impeller to bag diameter ratio of 0.38 and $\Delta z = 1.3d_2$. Based on this, the design criteria are inside the specifications for mammalian cell cultivations. Table 1 shows a detailed list of the geometrical dimensions for the CultiBag STR family (Fig. 1).

For the 50 and 200 L scale, the impellers are mounted on a rigid shaft. For scales greater or equal to 500 L, a telescopic shaft is used. Since the impeller shaft is connected to the motor via a magnetic coupling, the cultivation chamber remains close and sterile. For the CultiBag STR 50–1000 an axial and for the CultiBag STR 2000 L a radial magnetic coupling is used. There is the opportunity to select between the 3-blade-segment impeller with an axial flow pattern and 6-blade-disk impeller with a radial flow pattern. This study will focus on the process engineering characterization with the configuration 2 × 3-blade-segment impellers with a downward mixing. The configuration 6-blade-disk (bottom) and 3-blade-segment (top) impeller is also available for the CultiBag STR family. By a sparger installed at the lowest position of the bag, the stirred single-use bioreactor family is actively aerated. For the study the combi sparger was investigated, which has a ring sparger part (hole diameter = 0.8 mm) and a micro sparger part (hole diameter = 0.15 mm). The ring sparger part produces gas bubbles with a diameter of 5 mm [22]. Hence, there is an efficient CO₂ removal and an improved oxygen transfer by the micro sparger part due to smaller gas bubbles.

The impellers, sparger, and optochemical probes for pH- and pO₂-measurement are preinstalled. Sterile conditions of the bags are guaranteed by gamma irradiation. The cultivation chamber is installed into the bag holder made of stainless steel to increase the robustness. To ensure a constant temperature, the bag holder has a double wall. The height of the double wall corresponds to half of the maximal filling height (h_1) for the CultiBag STR

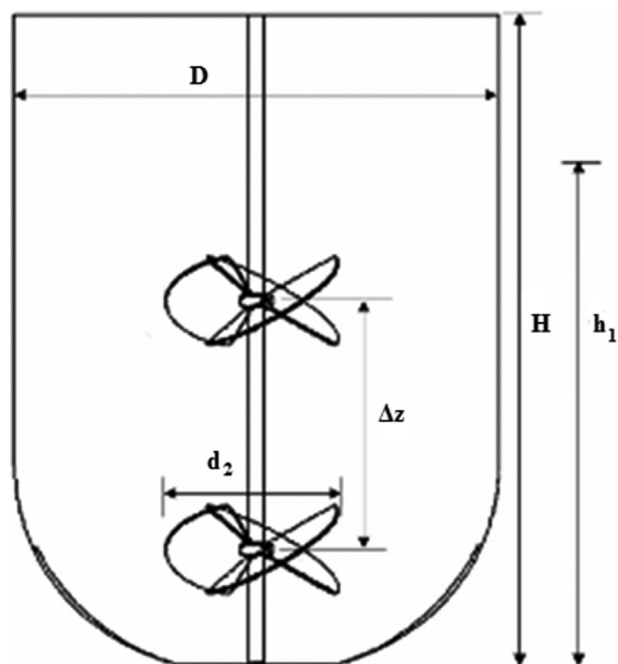


Figure 1. Schematics showing the key geometrical dimensions of a CultiBag STR bioreactor: D, bag diameter; d_2 , impeller diameter; H, bag height; h_1 , liquid height; Δz , distance between impellers.

50–1000 L. In order to increase the heat transfer of the CultiBag STR 2000 L, the height of the double wall was increased.

2.2 Determination of the power input per volume

To characterize the impeller types/configurations and to quantify P/V_L , the dimensionless Newton number (Ne) was determined. Therefore, torque measurements were carried out for the 50 and 200 L scale and Ne was determined by Eq. (3).

$$Ne = \frac{2 \cdot \pi \cdot M_t}{\rho \cdot N^2 \cdot d_2^5} \quad (3)$$

where Ne is the Newton number [–] and M_t is the torque [Nm]. For the measurements, a torque transducer with a measuring range up to 10 Nm (ETH-Messtechnik) was used. The single-use bioreactors were filled with RO water to their maximal filling volume. For the determination, only values under turbulent conditions were considered because a constant Ne is achieved. For the trials, P/V_L in an ungassed system was determined because the gassing rates for cell culture processes are generally low. Hence, the influence on the measurement can be neglected.

2.3 Mixing time determination

The mixing times were determined by the decolorization method, which is based on the color change of a dye [23]. 2 mL L^{–1} potassium iodide (stock solution containing: 40 g L^{–1} potassium iodide and 20 g L^{–1} iodine) and 5 mL L^{–1} starch solution (containing: 10 g L^{–1} starch) were added into the bioreactors filled with RO water to their maximal filling volume leading to a dark blue colorization. Afterwards, 4 mL L^{–1} sodium thiosulfate

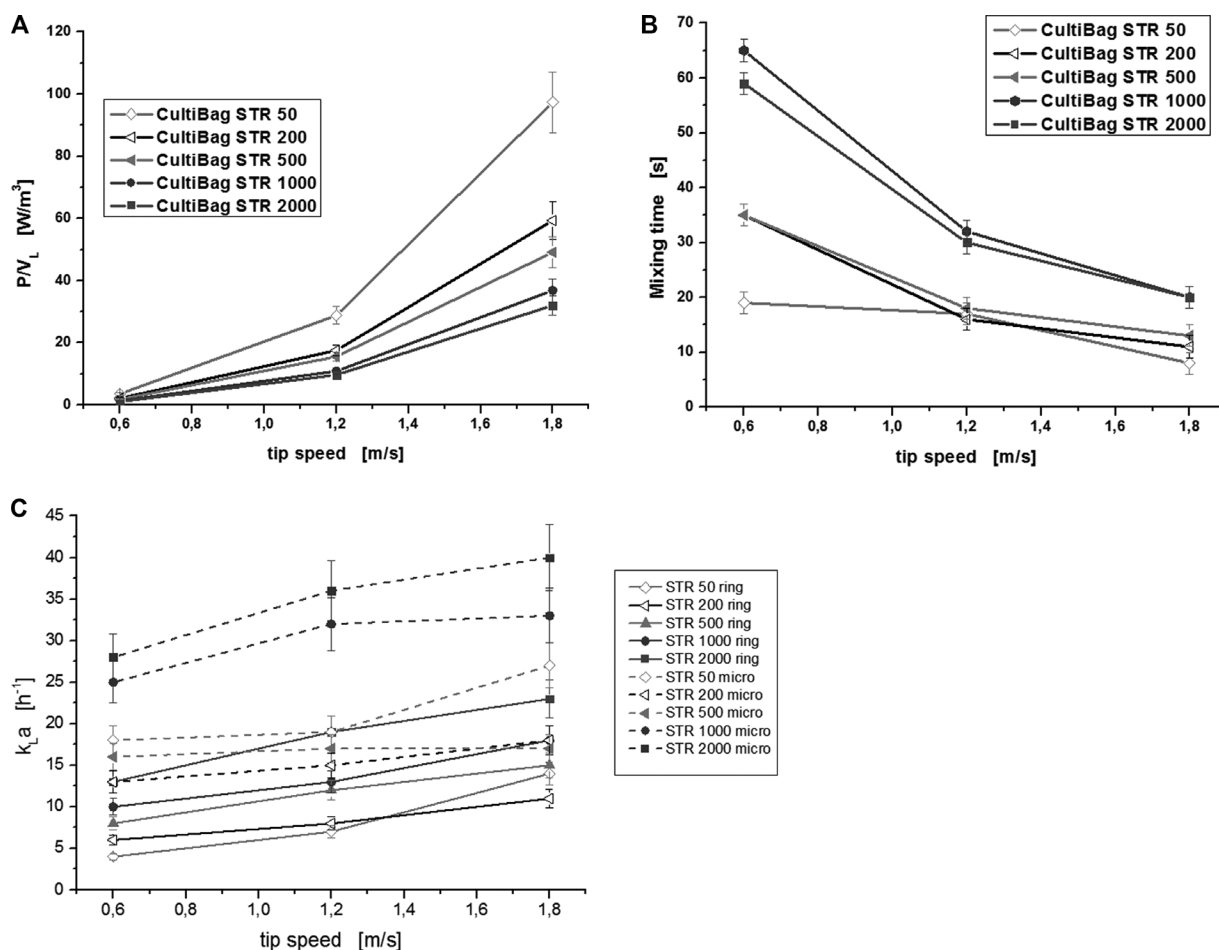


Figure 2. Results of the process engineering characterization of the CultiBag STR family from 50–2000 L scale with the configuration 2×3-blade-segment impeller. (A) Power input per volume, where a single determination was performed. Based on previous measurements, a standard deviation of $\pm 10\%$ was assumed. (B) Mixing times in dependence of the tip speed. Based on previous studies, a standard deviation of ± 2 s was assumed. (C) Shows the k_La values for the aeration with the ring and micro sparger part. Based on previous studies, a standard deviation of $\pm 10\%$ was assumed. The error bars show the standard deviation, which was calculated from triplicates of prior experiments.

(Na₂S₂O₂) solution with 0.1 M was added, which decolorizes the solution. The time between Na₂S₂O₂ addition to complete decolorization (goodness of mixture = 1) is defined as the mixing time. The mixing time is determined by visual judgment, hence it is a subjective procedure and can lead to deviations. An advantage is that possible zones of poor mixing are detectable.

2.4 Volumetric mass transfer coefficient (k_La) measurements

For the quantification of the oxygen transfer rate of the bioreactor family, the k_La value was determined by the gassing out method [24]. The bags were filled with 1X PBS buffer, and the liquid was deoxygenated by the supplementation of nitrogen to achieve a pO₂ below 10%. After N₂ addition air was sparged and the pO₂ increase was recorded. All measurements were carried out at room temperature. The k_La was calculated from the slope of the mass balance (see Eq. (4)).

$$\ln\left(\frac{C^* - C_1}{C^* - C_2}\right) = k_La \cdot (t_2 - t_1) \quad (4)$$

where C_1 and C_2 [mmol_{O₂} L⁻¹] are the dissolved oxygen concentrations at time t_1 and t_2 [h]. For the measurements, a preinstalled optochemical probe (PreSens) was used. All measurements were performed at a gas flow rate of 0.1 vvm.

3 Results

To characterize P/V_L , the Newton number was determined (see Section 2.2). For the 2×3-blade-segment impeller, a Ne of ~ 1.3 was calculated for the scales 50 and 200 L. Due to geometrical similarities, it is suggested that the other scales have a same Ne . Because for all scales $Re > 10,000$, turbulent flow conditions are present [25] and it can be concluded that Ne is constant for the tip speeds between 0.6–1.8 m/s. The power input per volume (P/V_L) is an important process engineering parameter

and was calculated based on the Newton number (Eq. (5)) (see Fig. 2A).

$$\frac{P}{V_L} = \frac{Ne \cdot \rho \cdot N^3 \cdot d_t^5}{V_L} \quad (5)$$

where P/V_L is the power input per volume [W m^{-3}] and V_L is the filling volume [m^3]. P/V_L increases with the tip speed and decrease with the scale. For example P/V_L increases for the CultiBag STR 50 L from 4 W m^{-3} at a tip speed of 0.6 m s^{-1} to 97 W m^{-3} at $u_{\text{tip}} = 1.8 \text{ m s}^{-1}$. P/V_L is for the CultiBag STR 2000 L three times lower compared to the CultiBag STR 50 L at the same tip speeds. As a consequence, to achieve the same P/V_L in larger scales, the tip speed has to be increased.

Shear forces correlate directly with P/V_L because the generation of micro eddies is influenced. If these eddies are of the same magnitude like the diameter of the cultivated organisms, the cell walls can be disrupted [14]. The size of the eddies can be calculated with the kinematic viscosity and P/V_L (see Eq. (6)).

$$l = \left(\frac{\nu^3}{P/V_L} \right)^{\frac{1}{4}} \quad (6)$$

where l is the diameter of the eddies [m] and ν is the kinematic viscosity [$\text{m}^2 \text{ s}^{-1}$] (ν of water was assumed). For the P/V_L described above, an eddy size between 56 and $178 \text{ }\mu\text{m}$ is achieved. Mammalian cells like CHO have a diameter of $10\text{--}20 \text{ }\mu\text{m}$ [26] and consequently cell damage can be excluded.

For the verification of the mixing efficiency, the mixing times were determined as a function of the tip speed (see Fig. 2B). It is obvious that the mixing time decreased by increasing tip speed resulting from the higher P/V_L , and higher mixing times were determined for increasing scales. The CultiBag STR 50 L had at $u_{\text{tip}} = 0.6 \text{ m s}^{-1}$, a mixing time of 19 s and at $u_{\text{tip}} = 1.8 \text{ m s}^{-1}$ of 8 s . The lowest mixing time which could be achieved for the CultiBag STR 2000 L was 20 s .

Figure 2C shows the characteristics of the $k_L a$ value for different tip speeds and scales. For the trials the influence of the ring and micro sparger part was investigated. The $k_L a$ values increase with tip speed independent of the scale and configuration. This can be explained by the increased P/V_L . It was observed that the $k_L a$ value is higher for larger scales. The reason is presumably the increased liquid height and the associated longer residence time of the gas bubbles resulting in a more efficient oxygen transfer [12]. This is also indicated by a linear correlation between the filling height and the $k_L a$ value. For $u_{\text{tip}} = 0.6 \text{ m s}^{-1}$ and e.g. ring sparger aeration a dependency of $k_L a = 0.077 h_L$ is given ($R^2 = 0.99$).

Volumetric mass transfer coefficients of 23 h^{-1} can be achieved for the CultiBag STR 2000 L using ring sparger aeration. Due to the smaller gas bubbles, the $k_L a$ achieved by the aeration with micro sparger part is higher and the $k_L a$ values are approximately doubled. The results demonstrate that oxygen limitations can be excluded during a process scale-up because the oxygen transfer rate is higher for larger scales.

Table 2. Alterations of the process engineering parameters for the scale-up from the CultiBag STR 50 to the CultiBag STR 2000. The initial CHO process in the CultiBag STR 50 was performed at a stirrer speed of 150 rpm [27].

	N [rpm]	u_{tip} [m/s]	Re [–]	P/V_L [W m^{-3}]
Process engineering parameters CultiBag STR 50	150	1.1	49,420	22.3
Equal N for CultiBag STR 2000	150	3.9	605,406	293.1
Equal u_{tip} for CultiBag STR 2000	42	1.1	171,936	6.7
Equal Re for CultiBag STR 2000	12	0.31	49,420	0.15
Equal P/V_L for CultiBag STR 2000	63	1.6	254,270	22.3

4 Concluding remarks

The main application of the presented single-use bioreactor family is the cultivation of mammalian cells. To verify if this bioreactor family guarantees optimal growth conditions for mammalian cells, different process engineering parameters were determined. With a P/V_L below 100 W m^{-3} the bioreactor can be regarded as suitable for mammalian cell lines. The generated micro eddies have a significantly larger diameter than animal cells, and therefore damaging can be excluded. For all scales, mixing times below 30 s can be achieved. This is comparable to stainless steel bioreactors used for cultivation of mammalian cells [10].

A further critical parameter, especially for high cell density cultivations, is the oxygen transfer rate. A modern CHO process with a peak cell density of $27\text{--}28 \times 10^6 \text{ cells mL}^{-1}$ had an OUR of 7 mmol (Lh)^{-1} [27]. Therefore, a $k_L a > 7 \text{ h}^{-1}$ is needed when pure oxygen aeration is assumed, which is the case for all scales.

Besides the feasibility for mammalian cell line cultivation, care was taken regarding the scalability of the single-use bioreactor family. Regarding the possibility for scale-up and scale-down, the fact that the design directly relates to those of stainless steel bioreactors represents a large advantage. Care was taken that the geometrical properties of the cultivation chamber and impellers are similar for all scales. This similarity is a prerequisite for a successful scale-up/scale-down or process transfer.

To illustrate a possible process transfer a scale-up matrix is given in Table 2. A CHO process, which was performed at a stirrer speed of 150 rpm in the CultiBag STR 50 L, was assumed [27], and the alterations of the process engineering parameters in the CultiBag STR 2000 L are listed. If the stirrer speed is selected as scale-up criterion, the tip speed is increased to 3.86 m s^{-1} , which is outside the specifications for mammalian cell cultivations. This can result in damaging of the cells due to shear forces, which is

also evident by P/V_L which is above the proposed values for mammalian cells. When u_{tip} remains constant, the stirrer speed and P/V_L decreases. This results in a decrease of the mixing efficiency and $k_L a$ value, and consequently limitation can occur. The same phenomena in an enhanced way can be concluded if Re is kept constant.

For the use of P/V_L as scale-up criterion, the tip speed is slightly increased, and the stirrer speed decreases making cell damage unlikely. An insufficient mixing can be excluded if the process is performed at $u_{tip} = 0.6$ – 1.8 m/s because the mixing times are below the specifications [10]. Furthermore, oxygen limitations are not possible when the same cultivation process is used because the $k_L a$ is higher for larger scales. Based on the results, P/V_L is recommended when a scale-up is performed for the CultiBag STR family because the changes of the process engineering parameters are in good agreement with the key process parameters for mammalian cell cultivations.

Practical application

This study deals with the process engineering characterization of the stirred single-use bioreactor family BIOSTAT® STR from 50–2000 L scale. For the characterization, parameters commonly used as scale-up criteria like the power input per volume, mixing time, and volumetric mass transfer coefficient ($k_L a$) were measured. Furthermore, the dimensionless Newton number was determined to describe the system properties. These parameters were compared to the demands of modern mammalian cell cultivations, which define the design space. Hence, a conclusion about the suitability for cell culture applications can be made. Such a complete characterization study of a single-use bioreactor family across the scales will therefore significantly facilitate the scale-up and process transfers of processes based on these technologies.

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The authors have declared no conflict of interest.

Nomenclature

A	$[m^2 m^{-3}]$	Specific transfer area
C	$[mol L^{-1}]$	Dissolved oxygen concentration
C^*	$[mol L^{-1}]$	Oxygen saturation concentration
C_1	$[mol L^{-1}]$	Dissolved oxygen concentration at time t_1
C_2	$[mol L^{-1}]$	Dissolved oxygen concentration at time t_2
d_2	$[m]$	Impeller diameter
D	$[m]$	Bag diameter
h_1	$[m]$	Liquid height
H	$[m]$	Bag height
$H:D$	$[-]$	Height to diameter ratio
$k_L a$	$[h^{-1}]$	Volumetric mass transfer coefficient

L	$[m]$	Eddy diameter
M_t	$[Nm]$	Torque
N	$[s^{-1}]$	Stirrer speed
Ne	$[-]$	Newton number
OUR	$[mol(Lh)^{-1}]$	Oxygen uptake rate
OTR	$[mol(Lh)^{-1}]$	Oxygen transfer rate
pO_2	$[\%]$	Oxygen partial pressure
P/V_L	$[W m^{-3}]$	Power input per volume
Re	$[-]$	Reynolds number
u_{tip}	$[m s^{-1}]$	Tip speed
V_L	$[L]$	Filling volume
X	$[cells mL^{-1}]$	Cell density
Y_{X/O_2}	$[cells/mmol O_2]$	Oxygen yield coefficient
Δz	$[m]$	Distance between impellers
μ	$[h^{-1}]$	Specific growth rate
N	$[m^2 s^{-1}]$	Kinematic viscosity
P	$[kg m^{-3}]$	Density

5 References

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